STUDENT PERFORMANCE ANALYTICS SYSTEM

Project Documentation | Python with AI Internship

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Training organization	EWB EDU Private Limited (EWB Courses)
Technologies	Python, NumPy, Pandas, CSV
Dataset	20 sample student records
GitHub repository	https://github.com/YashSharmaECE/student-performance-analytics

This document describes the project objective, dataset, implementation, analysis results, and evidence captured from the program running in VS Code.

1. Project Objective

The Student Performance Analytics System reads student data from a CSV file and uses Python, Pandas, and NumPy to calculate student-wise totals and averages, assign grades, determine pass/fail status, summarize class performance, analyze subject averages, identify the best-performing subject, and list the top five performers.

2. Technologies Used

Python — core programming and logic; Pandas — loading, processing, and exporting tabular data; NumPy — numerical calculations such as averages, maximums, and minimums; CSV — input dataset and generated results report.

3. Dataset Description

The input file students.csv contains 20 sample student records. It includes student ID, name, department, marks in Python, Mathematics, AI, Electronics, and English, plus attendance. The program generates student_results.csv with calculated Total, Average, Grade, and Result fields.

4. Implementation

1. Load the input CSV into a Pandas DataFrame.
2. Process each student's subject marks using reusable functions in functions.py.
3. Calculate total and average marks, assign a grade, and check the result condition.
4. Display the student-wise report and class statistics.
5. Calculate subject-wise averages and identify the best-performing subject.
6. Sort records by average marks and display the top five performers.
7. Export the complete processed DataFrame to student_results.csv.

5. Key Features

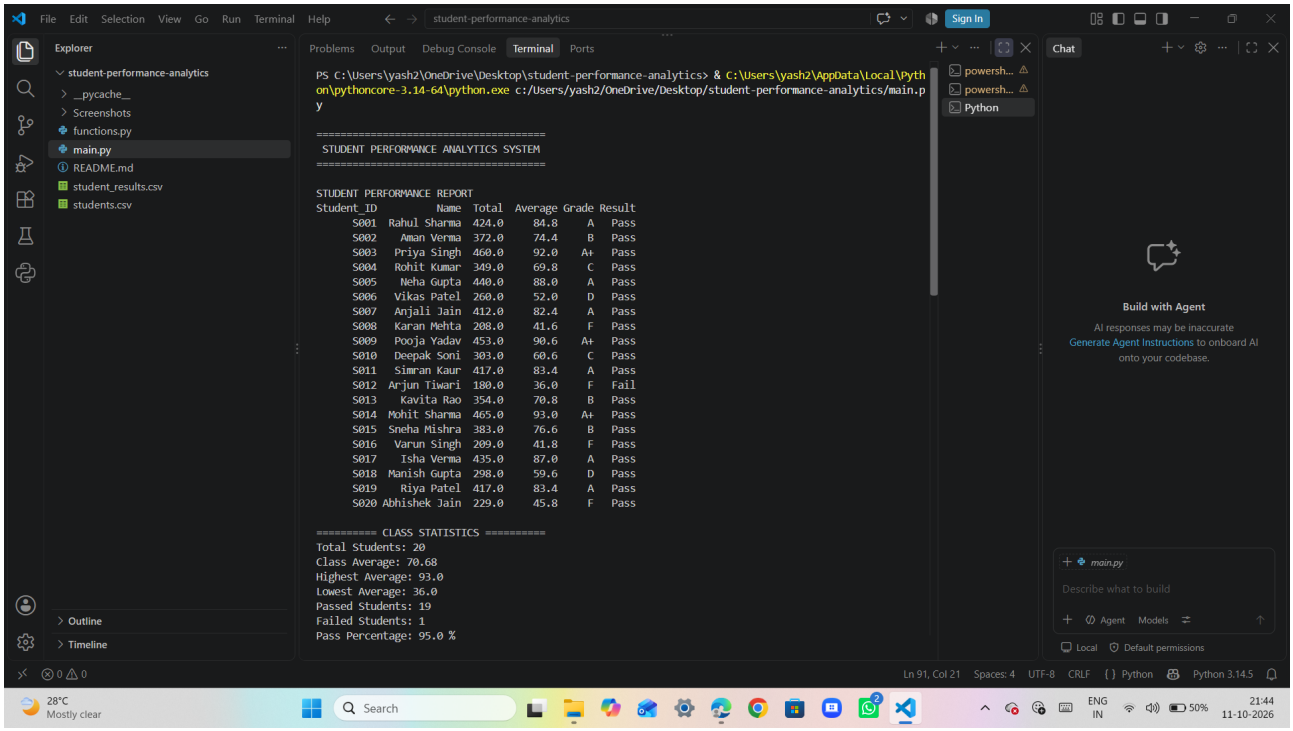
Student-wise totals, averages, grades and results; class average, highest and lowest average; pass/fail count and pass percentage; subject-wise average analysis; best-performing subject; top five performers; CSV export.

6. Result Summary

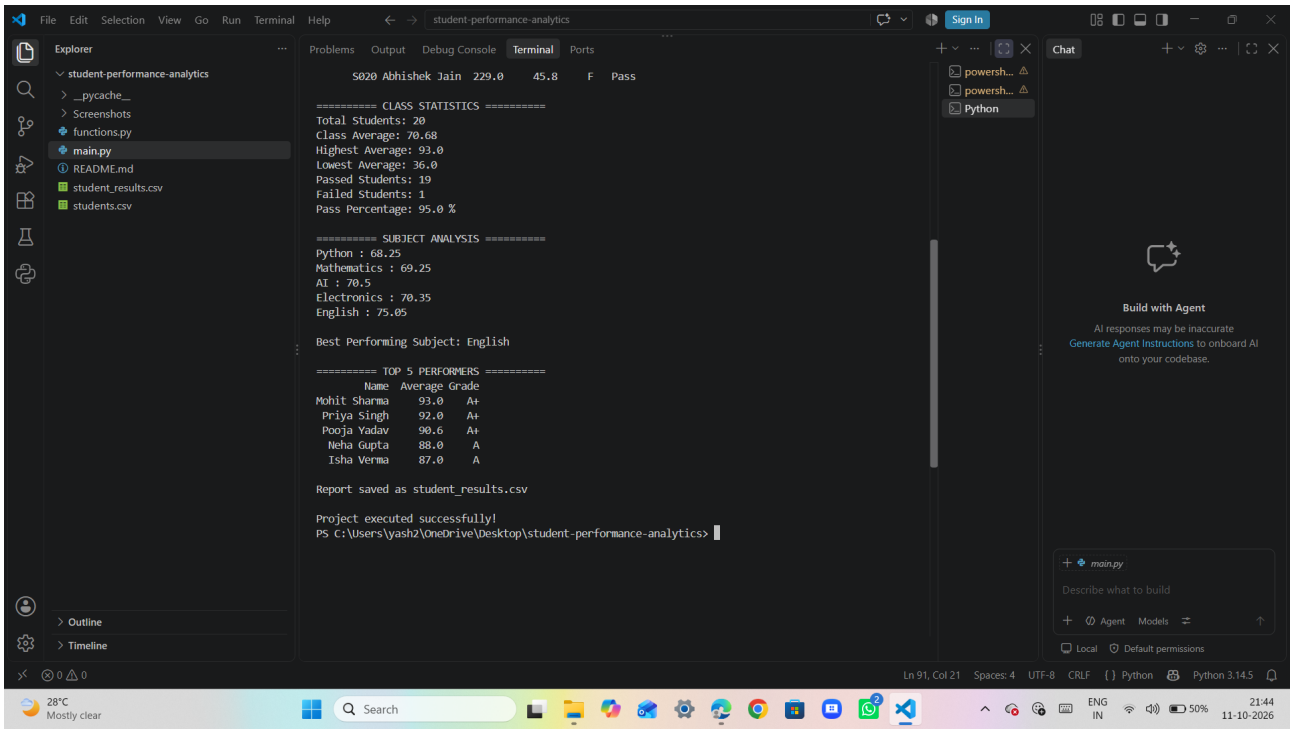
The captured run shows 20 students, a class average of 70.68, a highest average of 93.0, a lowest average of 36.0, 19 passed students, 1 failed student, and a pass percentage of 95.0%. The subject analysis identifies English as the best-performing subject in this sample run.

7. Actual Program Output

The screenshots below were captured from the program running in VS Code and show the full report, class statistics, subject averages, and top performers.



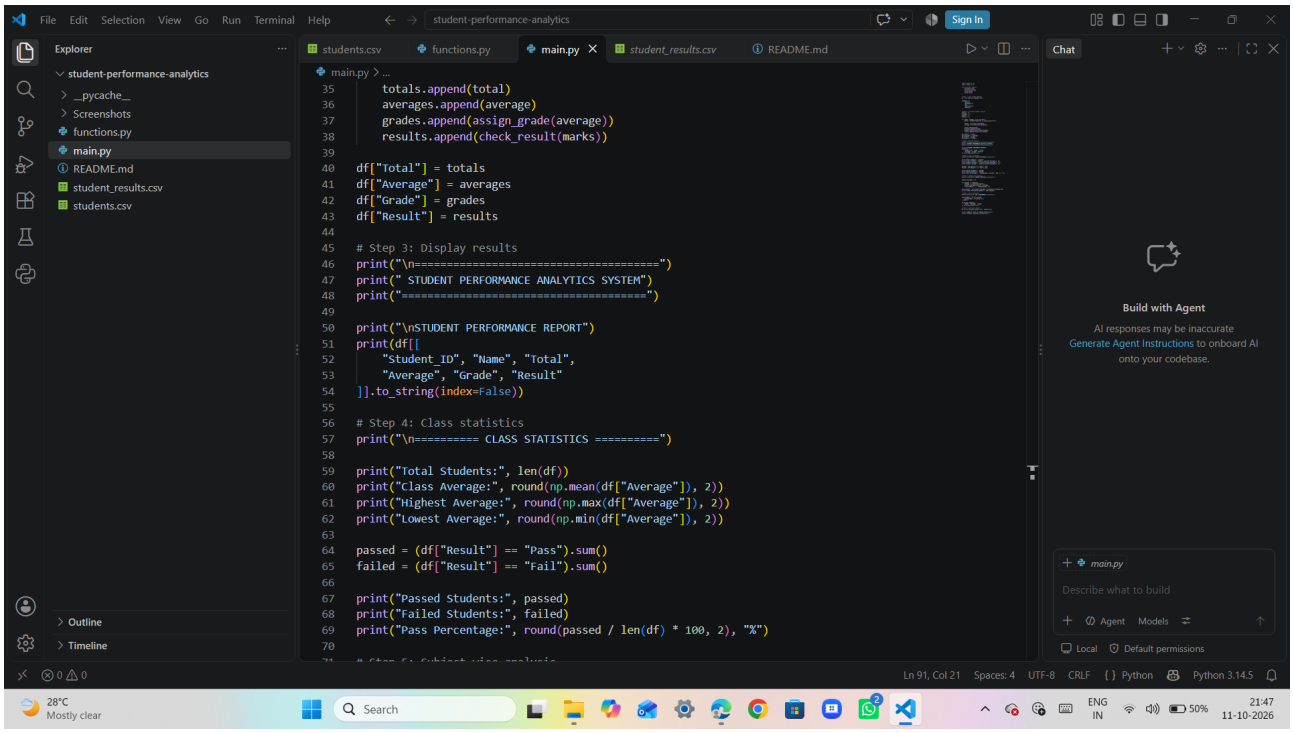
8. Subject Analysis and Top Performers



The program lists the top five students by average marks. The full student report is available in the generated CSV file.

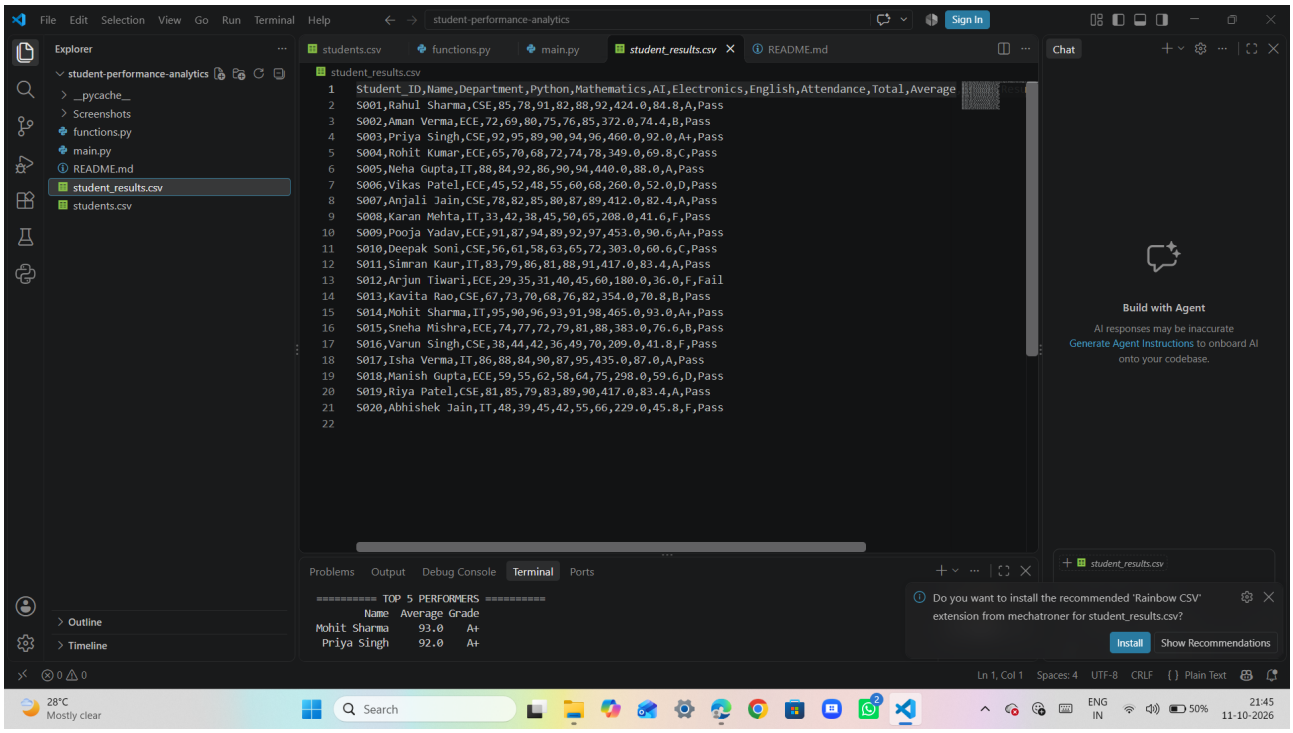
9. Code Evidence

The code uses Pandas and NumPy for data processing and imports reusable functions from functions.py.



10. Generated Results File

student_results.csv contains all 20 student records, with calculated Total, Average, Grade, and Result columns.



11. How to Run the Project

1. Install Python.
2. Install dependencies: **python -m pip install numpy pandas**.
3. Keep main.py, functions.py, and students.csv in the same folder.
4. Open a terminal in that folder and run **python main.py**.
5. The program displays analysis and creates student_results.csv.

12. Learning Outcomes

The project provides practice in reading CSV files, using Pandas DataFrames, applying NumPy calculations, writing reusable functions, iterating through records, using conditions, sorting results, summarizing data, and exporting processed data.

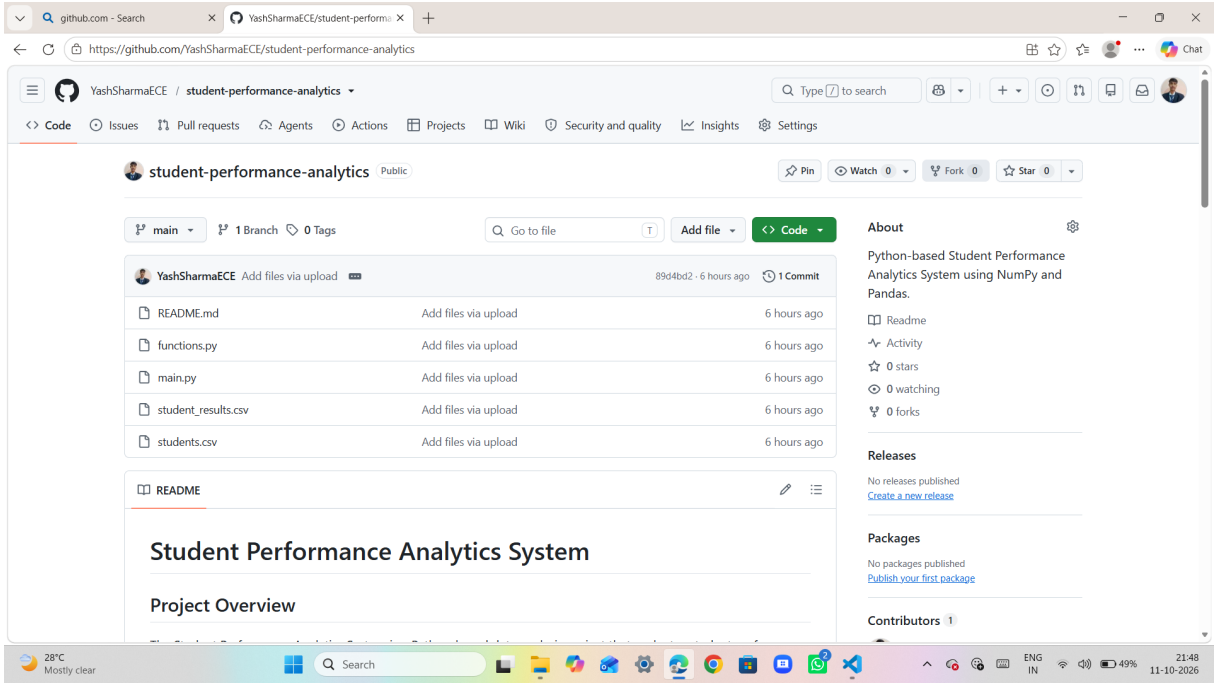
13. Project Files

main.py — main analysis program
functions.py — calculation and result functions
students.csv — input dataset
student_results.csv — generated report
README.md — project overview
Documentation PDF — project report

14. GitHub Repository

https://github.com/YashSharmaECE/student-performance-analytics

Repository evidence



Conclusion

The project successfully processes a sample student dataset and produces a student-wise report along with class-level and subject-level statistics. The generated CSV and actual execution screenshots demonstrate the workflow.